

More Than an Analogy: The Real Medium Under the Acoustic Metric

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Preamble — what this article does and does not claim

1. **This grounds analogue gravity’s own open question, not a new derivation.** Analogue gravity (Unruh 1981; Barceló–Liberati–Visser, “Analogue Gravity,” gr-qc/0505065) shows that excitations in a moving medium obey a curved Lorentzian “acoustic metric,” reproducing horizons and Hawking radiation. Its authors repeatedly ask whether it can be “*more than an analogy*” — whether real spacetime could actually *be* the effective acoustic metric of some underlying medium — and state plainly (§7.8) that “*no specific and compelling model of the relevant microphysics has yet emerged.*” This article argues Event Density (ED) is a candidate for that microphysics. ED’s acoustic metric is already derived elsewhere in the corpus; it is assembled here, not re-derived.
2. **ED’s acoustic metric is kinematic and non-Einstein — this is *parity* with analogue gravity, not a defect.** No known emergent-gravity model reproduces the Einstein equations from its microdynamics; even Volovik’s best route (superfluid ^3He) yields fluid-mechanical or Nordström dynamics, not Einstein (BLV §7.9), and “the Einstein equations have to date not been reproduced in any known condensed-matter system” (§7.4). ED reaches a *khronometric, weak-field-Einstein* dynamics by its own separate routes, which is at or slightly past that frontier, but **not** full non-linear Einstein. Curvature emergence is ED’s open bridge.
3. **The Hawking-analogue results are tiered, and generic ED horizons are cold.** ED’s $T=\kappa/2\pi$ and Planck spectrum are FORCED-CONDITIONAL on the Euclidean-continuation assumption, with coefficients inherited (arc-Hawking); and a *generic* ED interior horizon is acoustically extremal ($\kappa=0$, $T_H=0$) — ED does not automatically radiate. Any “ED reproduces analogue Hawking radiation” carries these caveats.
4. **ED dissolves the cosmological-constant *catastrophe*, and inherits only Λ ’s *value*.** The notorious $\sim M_P^4$ overshoot comes from summing an infinite tower of zero-point modes; ED’s substrate is discrete, so there is no infinite tower and that term is *structurally absent* (the report’s “dissolved catastrophe”; Paper_038_5; UV-finiteness Paper_040, Paper_111). ED identifies Λ ’s form ($\rho_\Lambda = (3/8\pi)H_0^2 M_P^2$) and reframes its smallness as the Friedmann ratio $(H_0/M_P)^2 \approx 10^{-122}$, two ordinary scales rather than a fine-tuning. What remains inherited is Λ ’s *value* via H_0 (deriving H_0 , Route-A, is open). This is a *different* mechanism than the “liquid-equilibrium” CC resolution BLV note (§7.10), reaching the same place.
5. **The claim is not that a laboratory analogue *is* gravity.** BLV explicitly reject the idea that “manipulating an analogue spacetime metric will lead to real physical changes in the gravitational field” (their footnote 2), and so does this article. ED’s claim is narrower and different: the ED

substrate is a candidate real medium beneath *real* spacetime, of which the metric is the coarse-grained acoustic reading.

6. **ED's results are conditional derivations from thirteen posited primitives**, never derivations from nothing, and the framework is not experimentally confirmed. Tiers are marked per claim.
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1. What Event Density is, and the family this belongs to

Event Density is a discrete, relational substrate: a participation graph whose channels carry a bandwidth $b \geq 0$ and a phase, with one process primitive, **commitment** (P11), the irreversible act that writes the arrow of time into the law. Gravity is emergent — the metric is the *reading* of the bandwidth field's connectivity ($g \sim 1/b$, Paper_GR-I) — and its gravitational class is **khronometric**: Einstein's tensor modes plus one scalar, the khronon, the arrow made dynamical (Paper_GR-II).

This is the fifth paper in the *microscopic-completions* family. Its companions treat the *dynamics*-side emergent-gravity results — Jacobson's equation of state, Padmanabhan's equipartition, the khronometric Clausius derivation, and Sakharov's induced action. This one treats the *kinematics*-side and ontology result: the acoustic metric, and analogue gravity's standing question of whether there is a real medium beneath it.

2. Analogue gravity, and the question it leaves open

Analogue gravity begins with a concrete fact: sound in a moving fluid obeys the same wave equation as a scalar field in a curved Lorentzian geometry — the **acoustic metric**, built from the fluid's density and flow. A supersonic flow makes a “dumb hole,” the acoustic analogue of a black hole, complete with a horizon and (where a second quantization is meaningful) phononic Hawking radiation. This has become a mature experimental program (BEC, optical, water-tank analogues).

But the field hedges the construction as an *analogy*, and names precisely where it stops:

- **The dynamics is missing.** “It is a well-known issue that the expected relativistic dynamics, i.e., the Einstein equations, have to date not been reproduced in any known condensed-matter system” (BLV §7.4). Analogue models supply the metric's *kinematics* — the geometry excitations propagate on — but the fluid never satisfies Einstein's equations as a consequence of its own microdynamics.
- **The microphysics is unspecified.** Discussing emergent gravity, BLV write that “it is plausible (even though **no specific and compelling model of the relevant microphysics has yet emerged**) that the spacetime manifold and spacetime metric might arise only once one averages over suitable microphysical degrees of freedom” (§7.8).
- **The medium is disowned.** The whole construction is presented as an analogy precisely because there is no claimed *real* medium under real spacetime; the acoustic metric is the effective geometry of a *stand-in* fluid.

So analogue gravity's own open question, in its authors' words, is whether it can be “more than an analogy”: whether there is a real medium whose coarse-graining *is* spacetime, and a specific microphysics to name it. The rest of this article proposes ED as an answer, at the tier the field itself operates.

3. ED's emergent metric is an acoustic metric — and its medium is real

ED does not presuppose a fluid; it has a substrate, and its metric is that substrate's acoustic metric in the exact technical sense of analogue gravity:

- **The acoustic metric is derived — in two forms worth keeping distinct.** (a) ED's *literal* Unruh-Visser acoustic metric: the effective line element $g^{\text{eff}} = \text{diag}(-M(\rho_0), 1, 1, 1)$, sound speed $c_s = \sqrt{M(\rho_0)}$, from the usual conformal field redefinition, flat spatial section, acoustic horizon at the mobility-collapse surface $M \rightarrow 0$. This is genuinely *kinematic* (the metric is slaved to ρ_0) and at parity with a standard analogue metric. (The working memo flags its particular $c_s = \sqrt{M_0} \approx 1.67c$ as not literally viable; this does not touch the parity point, since analogue metrics generically carry $c_s \neq c$.) (b) Paper_GR-I's weak-field metric $g \sim 1/b(g_{rr} \sim b^{-1}, g_{\theta\theta}g_{\phi\phi}) = -1$, factor-of-two bending), which has a *curved* spatial sector and is *dynamical* — the fixed point of ED's bandwidth rule $\nabla^2 b \sim \rho$ (GR-III), not kinematic. Both are acoustic-class emergent metrics; (a) carries the kinematic-parity claim, (b) the weak-field-Einstein step of §6. Either way ED's metric is an acoustic metric, at derivation tier.
- **The medium is the substrate, and it is not a stand-in.** Where the analogue program's medium is a laboratory fluid deputizing for spacetime, ED's medium is the participation substrate itself — the thing the corpus posits to be beneath *real* spacetime. As the foundations paper already states, “ED's substrate is not a fluid in the standard hydrodynamic sense but a participation-channel network; the acoustic-metric emergence is structurally common, the substrate identification is ED-specific.” This is the ontological upgrade: ED promotes the acoustic metric from *analogy* to *ontology* by supplying a real medium.

The caution, honored. This is emphatically not the claim BLV reject — that tuning a lab analogue alters real gravity. ED says the *substrate* is the real medium beneath spacetime, and the metric is its coarse-grained acoustic reading; it says nothing about laboratory fluids being gravity. The ED lab predictions that do exist are, honestly, “acoustic-metric kinematic, shared with standard analogue gravity” — a parity test, not a claim that the analogue is the real thing.

4. The dynamical aether: ED supplies what analogue models lack

Here is the sharpest point of contact, and it is one BLV name explicitly. In §7.3 they observe that a nontrivial (Lorentz-violating) dispersion relation implicitly introduces an **aether field** — a preferred rest frame V^a — so that analogue models are a *kinematic* implementation of Einstein-aether theory. And they flag the gap:

“in standard analogue models such an aether field does not come with its own dynamics: It is a background structure... However, in a gravitation theory context one might still want to require background independence... In this case one has to **provide the aether field with a suitable dynamics**.” (BLV §7.3)

ED provides exactly that dynamics. ED's preferred frame is not a background structure imposed by a laboratory; it is the khronon — the arrow of commitment (P11) made geometric and *dynamical* (Paper_GR-II): a hypersurface-orthogonal Einstein-aether field that genuinely *propagates* (a scalar mode, not a background), of khronometric class. (Its explicit field-equation coefficients are F -dependent and deferred, Paper_GR-II §5, §8; what is established is the *class* and the mode's dynamical, propagating character — which is exactly the kinematic-vs-dynamical distinction at issue. That ED's dispersion rest-frame and its foliation/arrow are the *same* field is an ED-internal

identification, GR-II’s single-cone single-substrate claim, not something analogue gravity establishes.) So where analogue gravity carries a *kinematic* aether with no dynamics, ED carries the *dynamical* aether the gravitational context requires. The companion papers show that this dynamical aether is not a liability: its preferred-frame footprint is ~ 70 orders suppressed by sparse becoming (Paper_GR-IV), and it is exactly what lets ED run Jacobson’s equation-of-state derivation cleanly (Paper_KhronometricEquationOfState). Analogue gravity’s kinematic aether is a background; ED’s is the arrow, and it moves.

5. The standard obstructions to emergent gravity do not bite ED

Emergent-gravity programs face two well-known theorems. Read carefully (as BLV do), neither obstructs ED:

- **Weinberg-Witten (§7.6).** Often cited as forbidding emergent massless spin-two, but BLV note “at the kinematic level it has nothing to say,” and at the dynamic level it “is derived under specific technical assumptions (strict Lorentz invariance in flat spacetime) that are not applicable” to Lorentz-violating analogue/emergent settings. Weinberg and Witten *themselves* add that the theorem “clearly does not apply to theories in which the gravitational field is a basic degree of freedom but the Einstein action is induced by quantum effects.” ED is doubly exempt: it works at the kinematic-metric level and it violates strict Lorentz invariance (khronometric), and its companion Sakharov paper is precisely the induced-action case W-W exclude.
- **Marolf (§7.7).** Argues emergent geometrodynamics needs kinematic non-locality — but the conclusion “rests on the restrictive premise that all observables in the UV completion must be gauge invariant... if gauge symmetries are also emergent, locality could be preserved.” ED’s diffeomorphism invariance is itself *emergent* (a continuum-limit feature, reduced to foliation-preserving form, Paper_GR-II), so it sits in exactly the class Marolf’s own caveat — that the gauge symmetries may themselves be emergent — exempts.

This is a “does not bite” argument, not a proof that ED succeeds; the point is only that the usual barriers erected against emergent gravity do not stand in ED’s way, for reasons internal to what ED is (kinematic metric, Lorentz-violating, emergent gauge structure).

6. Honest parity: ED sits at the field’s frontier, not past it

The temptation is to claim ED *solves* what analogue gravity could not. It does not, and the honest picture is better told as parity plus one step:

- **The dynamics.** BLV §7.9: even the most-developed route (Volovik’s ^3He) fails to reach Einstein — “the dynamics of the gravitational degrees of freedom is not Einstein, but of fluid-mechanical type,” and the best relativistic emergent case reaches only *Nordström* (spin-0) gravity, not Einstein. ED reaches a **khronometric, weak-field-Einstein** dynamics through its own routes (the bandwidth rule Paper_GR-III; the Jacobson/khronometric equation of state), recovering the weak-field Einstein metric and factor-of-two bending via an *emergent dynamical rule*, genuinely past Nordström. But the “one step” is bounded, and its ledger is stated: G and the coefficients are value-inherited (Paper_027), the Einstein-over-Nordström branch is selected by the labeled P-Commitment-Linear premise, and the geodesic identity is reduced not proven (Paper_GR-III). So the step is at the level of the *recovered weak-field metric and its tests*, not a coefficient-complete micro-derivation, and ED does **not** reach full nonlinear Einstein from the microdynamics. Deriving the curved dynamics and its coefficients from the raw graph is ED’s open curvature-

emergence program. ED reaches the frontier the field defines, one bounded step further, and stops where the field stops.

- **Information.** BLV §7.11: in an emergent-gravity picture, black-hole information loss “could be a mere artifact of the classical treatment,” unitary on the full Hilbert space of “spacetime atoms plus Hawking quanta.” ED’s own horizon/information result (ED-10) is exactly this resolution with the atoms named: information is preserved by substrate recording, unitary on the full participation structure, non-unitary only in the mean-field (metric-only) reduction. ED supplies the “spacetime atoms” the analogue argument gestures at.
- **The cosmological constant.** BLV §7.10 note that an emergent (liquid-like) medium can force its vacuum energy small. ED reaches the same place by a *different* mechanism: its discreteness means there is no infinite mode-tower, so the $\sim M_P^4$ catastrophe is *dissolved* (structurally absent), Λ ’s form is identified, and its smallness is the Friedmann ratio $(H_0/M_P)^2$, not a fine-tuning (the report’s “dissolved catastrophe”; Paper_038_5). Only Λ ’s *value* (via H_0) is inherited; the residual hierarchy (deriving H_0 , Route-A) is open. So ED does not merely inherit the problem — it dissolves the catastrophe and inherits only the last measured scale.

7. Honest tiers

- **Derived (ED):** the acoustic metric (arc-acoustic working memo; Paper_GR-I weak-field, khronometric); the khronon as the dynamical Einstein-aether field (Paper_GR-II).
- **Grounded / assembled (literature + ED):** the identification of ED’s metric as an acoustic metric in the Unruh–Visser sense; ED as a candidate microphysics for the emergent-gravity question BLV leave open; the “obstructions do not bite” reading of Weinberg–Witten and Marolf (following BLV’s own analysis).
- **Parity, not surpassing:** ED matches analogue gravity’s kinematic-metric-plus-non-Einstein-dynamics status (BLV §7.9); its one step further is the weak-field Einstein recovery via its separate routes.
- **Tiered / conditional:** the Hawking-analogue results (FORCED-CONDITIONAL on Euclidean continuation, coefficients inherited); generic ED horizons are cold ($\kappa=0$).
- **Dissolved + partly inherited:** the cosmological-constant *catastrophe* is dissolved (discrete substrate, no infinite tower), Λ ’s form identified and smallness reframed as $(H_0/M_P)^2$; only Λ ’s *value* (via H_0) is inherited (Paper_038_5, the report’s “dissolved catastrophe”).
- **Open / inherited:** full nonlinear Einstein from the microdynamics (curvature emergence); the value of the grain; deriving H_0 (the Λ residual, Route-A).

8. Position statement

For over four decades analogue gravity has shown that a moving medium carries a curved acoustic metric, horizons, and Hawking radiation — and for just as long it has hedged this as an analogy, because it has no real medium under real spacetime and no specific microphysics to name, and because no medium’s microdynamics has ever reproduced the Einstein equations. Event Density answers the ontological half of that standing question: it supplies a real medium, the participation substrate, of which the metric is the coarse-grained acoustic reading, and it supplies the one thing analogue models structurally lack — a *dynamical* aether, the khronon, where analogue models carry only a kinematic one. The obstruction theorems raised against emergent gravity do not bite ED, for reasons internal to its being kinematic, Lorentz-violating, and emergently gauge-structured. What ED does *not* do is exceed the field’s frontier: its metric is kinematic and non-Einstein, its generic horizons are cold, and its Hawking results rest on the Euclidean route (though the cosmological-constant *catastrophe* it dis-

solves rather than inherits, keeping only Λ 's value). It reaches that frontier and one honest step past it (weak-field Einstein by its own routes), and it stops where the whole field stops — at the full nonlinear dynamics from the microphysics, which remains open. The claim, precisely, is this: **analogue gravity asks whether the acoustic metric can be more than an analogy; Event Density is a candidate for the “yes,” because it supplies the real medium and the moving aether the analogy was standing in for.** None of this is offered as confirmed physics; it is offered as the grounding of the open question analogue gravity has carried since 1981.

Cross-references

- Unruh, W. G., *Phys. Rev. Lett.* **46**, 1351 (1981); Visser, M., gr-qc/9712010 (acoustic black holes); Barceló, Liberati & Visser, “Analogue Gravity,” *Living Rev. Relativ.* (2005; upd. 2011/2024), gr-qc/0505065 (read: intro §1; §7.3 nontrivial dispersion = kinematic Einstein-aether, “provide the aether with dynamics”; §7.4 Einstein not reproduced in condensed matter; §7.6 Weinberg–Witten, §7.7 Marolf; §7.8 emergent gravity / “no compelling microphysics yet”; §7.9 Volovik route not Einstein; §7.10 cosmological constant; §7.11 information; §8 conclusions; footnote 2 the “not viable” caution).
- **ED acoustic metric / emergence:** event-density/arcs/arc-acoustic/ED_Effective_Acoustic_Metric.md (the Unruh–Visser construction in ED; c_s viability caveat §6E); physics-papers/gravity/Paper_GR-I_WeakFieldEinsteinMetric.md (g~1/b realized acoustic metric, weak-field dynamical); event-density/arcs/arc-acoustic/ED_Acoustic_Metric_Curvature.md (cold/extremal horizons, Rindler-not-Schwarzschild); event-density/papers/Substrate_Gravity_Foundations/Substrate_ (the “participation-channel network, not a fluid” statement); event-density/theory/ED_Geometry_Emergence_ ((ii)-grade achievable, (i)-grade not, under current axioms).
- **ED dynamical aether:** physics-papers/gravity/Paper_GR-II (khronometric class, the khronon), Paper_GR-IV_ArrowsAlibi.md (α_1 suppression), Paper_KhronometricEquationOfState.md (the khronon lets Jacobson run).
- **ED Hawking / information / Λ :** event-density/arcs/arc-Hawking/H-1, H-7 (Hawking-analogue, FORCED-CONDITIONAL); foundations/BH_Thermal2Pi_FromNearHorizonRindler.md; ED-10 horizon/information; physics-papers/gravity/Paper_038_5_Lambda_V1_Cosmological.md (Λ inherited).
- **Family companions:** Paper_GravityAsEquationOfState.md (Jacobson), Paper_GravityAsHorizonEquiparti (Padmanabhan), Paper_KhronometricEquationOfState.md (khronometric), Paper_InducedGravityArena.md (Sakharov); README.md.